

Pi Day Math Crossword Solutions

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ONE DOWN

Observe that, using partial fraction decomposition,

$$\begin{aligned}\frac{1}{n^2(n+1)^2} &= \left(\frac{1}{n(n+1)}\right)^2 \\ &= \left(\frac{1}{n} - \frac{1}{n+1}\right)^2 \\ &= \frac{1}{n^2} - \frac{2}{n(n+1)} + \frac{1}{(n+1)^2}.\end{aligned}$$

Hence,

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)^2} &= \sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{2}{n(n+1)} + \frac{1}{(n+1)^2}\right) \\ &= \sum_{n=1}^{\infty} \frac{1}{n^2} - 2 \sum_{n=1}^{\infty} \frac{1}{n(n+1)} + \sum_{n=1}^{\infty} \frac{1}{(n+1)^2}.\end{aligned}$$

First, we know

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

from Basel's Problem. Moreover, we can simplify the telescoping sum as follows:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1.$$

Finally,

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{(n+1)^2} &= \sum_{n=1}^{\infty} \frac{1}{n^2} - \frac{1}{1^2} \\ &= \frac{\pi^2}{6} - 1.\end{aligned}$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)^2} = \left(\frac{\pi^2}{6}\right) - 2(1) + \left(\frac{\pi^2}{6} - 1\right) = \frac{\pi^2}{3} - 3.$$

Since $a = 2$, $b = 3$, and $c = 3$, $a(b+c) = 2(3+3) = \mathbf{12}$.

THREE ACROSS

$$\frac{\pi}{2} = 2 \arctan\left(\frac{1}{4}\right) + \arctan(x).$$

$$\frac{\pi}{2} - 2 \arctan\left(\frac{1}{4}\right) = \arctan(x).$$

$$\tan\left(\frac{\pi}{2} - 2 \arctan\left(\frac{1}{4}\right)\right) = \tan(\arctan(x)).$$

$$\cot\left(2 \arctan\left(\frac{1}{4}\right)\right) = x.$$

$$x = \frac{\cot^2\left(\arctan\left(\frac{1}{4}\right)\right) - 1}{2\cot\left(\arctan\left(\frac{1}{4}\right)\right)} = \frac{15}{8}.$$

Since $a = 15$ and $b = 8$, $a + b = 15 + 8 = \mathbf{23}$.

TWO DOWN

$$a_1 = \lfloor \pi * a_0 \rfloor = \lfloor \pi * \pi \rfloor = \lfloor \approx 9.87 \rfloor = 9.$$

$$a_2 = \lfloor \pi * a_1 \rfloor = \lfloor \pi * 9 \rfloor = \lfloor \approx 28.27 \rfloor = 28.$$

$$a_3 = \lfloor \pi * a_2 \rfloor = \lfloor \pi * 28 \rfloor = \lfloor \approx 87.96 \rfloor = 87.$$

$$\lfloor \frac{a_3}{2} \rfloor = \lfloor \frac{87}{2} \rfloor = \lfloor 43.5 \rfloor = \mathbf{43}.$$

1. 1	2. 4
3. 2	3